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General description

This chapter provides a general description of the product MVB Virtual Terminal and PCNode:

Properties of the Multifunction Vehicle Bus
Applications of the product MVB Virtual Terminal and
PCNode

Software and hardware of the product MVB Virtual
Terminal and PCNode

Multifunction Vehicle Bus – overview

The Multifunction Vehicle Bus (MVB) is a serial communication bus for railway vehicles. It connects programmable equipment with each other and attaches directly to simple sensors and actuators. It is not intended for communication *between* vehicles – this is the domain of the train bus.

The MVB is standardised by IEC, TC9, WG 22 in the standard *Electric Traction Equipment - Train Communication Network*, Parts 1 to 3.

Properties of the MVB

The environment of train vehicles is very hostile for electronic equipment. Hence the MVB bears properties which are not needed for buses in the computer environment.

Magnetic fields	Magnetic fields are omnipresent in train vehicles. Hence the bus must operate securely in this environment.
Operating temperature	Train vehicles circulate all over the world. Hence the operating temperature can be as low as in a Siberian winter or as high as open sunshine in the Arabian desert.
Mechanical stability	Mechanical shocks are common to train vehicles. Both the cables themselves and the connectors must be absolutely safe for these situations. This applies also to the sensors, actuators and controllers on the bus.
Media and topology	Copper lines are intended for short distances and optical fibres for long distances. Using redundant lines allows modification of the bus topology without interruption. With copper lines a power line may be included to power the <i>bus terminators</i> and also devices on the bus.
Signals	The MVB uses a media independent method of signalling with <i>Manchester encoding</i> .
Bus management	Availability is increased, because mastership can be alternated between two or more bus administrators. <i>Mastership</i> is transferred from one bus master to another by a token frame.
Redundant layout	The MVB allows redundant layout with both twisted wire pairs and with optical fibre. This increases availability. You can limit redundancy to critical segments of the bus.

Access modes

The MVB offers two access modes to the communications medium:

- In periodic access mode, a predefined sequence of addresses are polled with a well-defined and deterministic period. This mode is intended for the transmission of periodic data.
- In sporadic access mode, devices announce an event. This mode is intended for transmission of message data. Sporadic traffic uses the time between the fixed, periodic phases.

Summary of properties.

Property	Characteristics	
Communications medium	Copper: twisted wire pair, biased RS 485	Optical fibres and active star coupler
Topography	Bus	Star
Extension	30m for RS 485 segment with 32 taps	2000m
Redundancy	Duplicated media: send on both, receive on one. Redundant masters with rotation	
Gross data rate	1.5 Mb/s	
Response Time	Typical 4µs, maximum 43µs	
Address space	4095 physical devices, 4095 logical ports, 8-bit station address for messages	
Frame size (useful data)	16, 32, 64, 128 and 256 bits	
Medium access control	Centralised master. Mastership transfer by token passing for back-up masters	
Bus management	background polling of devices, service messages, bus configuration	
Regular operation	Periodic polling for <i>process data</i>	Sporadic access for <i>message data</i> with <i>arbitration</i>
Application	Process variables	Remote procedure call
Logical link control	Broadcast of source-addressed process data	Slave-to-slave <i>datagrams</i>

Applications of the MVB

Due to its special properties (see details on the following pages) the MVB has many areas of application outside the domain of train vehicles. For example, the MVB is best suited for controlling high voltage switching bays or controlling power plants.

Multifunction Vehicle Bus – details

This section provides background information for system engineers evaluating the capabilities of the Multifunction Vehicle Bus.

This section also provides the necessary details to understand the various items used in the dialogs of Virtual Terminal.

For a recapitulation of bus communication in general see the appendix chapter.

Bus medium

Topology of the MVB

The Multifunction Vehicle Bus is the standard data carrier in locomotives, coaches or groups of vehicles not separated during normal operation. It provides both the interconnection of programmable equipment among themselves and the interconnection of programmable equipment with their sensors and actuators. MVB supports up to 4095 devices.

MVB physical layer

¹⁾The Multifunction Vehicle Bus uses three different media, which all operate at a unified speed of 1.5Mb/s.

- an electrical medium for short distances (20m), for instance to connect devices within racks and cabinets. This uses standard RS 485 transceivers.
- an electrical medium for middle distances (up to 200m) using a shielded, twisted wire pair and transformer coupling to provide galvanic separation,. This uses standard IEC 1158-2 transformers and transceivers.
- optical fibre for longer distances (up to 2000m) and critical environments, such as locomotives.

While the electrical media preferable have a bus structure, the optical medium usually has a star structure centred on an active or passive star coupler. The optical medium can also be laid out as a bus, if the number of devices is small enough to permit active or passive optical taps.

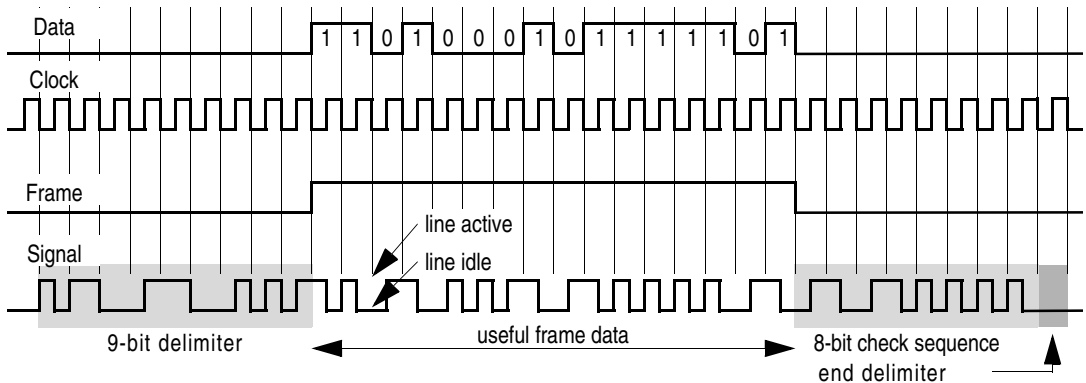
Configurations

In closed train sets, the MVB may span several vehicles. Fibre optics are not recommended for this purpose when the vehicles are frequently coupled and uncoupled.

The Multifunction Vehicle Bus has no provision to support configurable vehicles, the device addresses are fixed.

Signalling on the MVB

The data are encoded using a baseband Manchester code, which combines into one signal the data, the clock and the frame synchronisation. A “1” is transmitted as a negative transition (active to idle) in the middle of a bit cell, a “0” as a positive transition (idle to active)².



The frame data is headed by a 9-bit frame delimiter and followed an 8-bit check sequence.

Start delimiter

The start delimiter is used for synchronizing receivers and contains 3 Manchester code violations (transitions of 1.5 cell lengths) to distinguish it from valid data bit sequences. Master frames and slave frames have different delimiters to protect against synchronisation slips.

Check sequence

The check sequence is built over at most 64 data bits. It uses the high-safety TC57 telecontrol algorithm to provide a *Hamming distance* of 6. The Manchester encoding is inherently robust since both halves of a bit cell must be inverted to create a false bit. Taking this encoding into account, the achieved Hamming distance is even higher (around 8).

End delimiter

The frame is closed by an end delimiter to detect the end of a frame. For the fibre optics and the RS 485 transmission this is simply returning the line to the idle state for at least one bit time. In the middle-distance medium, however, the end delimiter has DC-free symmetrical shape.

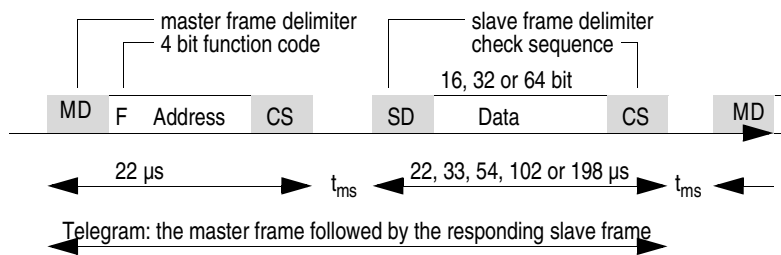
² The signal diagrams use “active on top” in [part 1 / p 42] and “idle on top” in [part 3 / p 37]!

MVB frames

- 3)The Multifunction Vehicle Bus uses two kinds of frames:
- The master frame, which is issued only by the bus master (administrator), and
 - The slave frame, which is sent by the source of the data in response to a master frame.

Telegram

A telegram consists of a master frame and the slave frame sent in response to it.



Reply time

4)The reply time t_{ms} is the delay between the end of a master frame and the beginning of the slave frame. This time consists of the sum of twice the propagation delays (line and regenerator), plus decoding and access delay at the slave. For example, the computed reply time is 4.36 μs for 30 m without repeater, and rises to 34 μs for 2 km lines and two repeater links in between. No configuration is allowed in which the expected reply time exceeds 42.7 μs, counted in the slowest line when redundant lines are used.

At the slave side, the response time between end of master frame and start of slave frame must be between 1.4 μs and 4 μs.

Function code

The master frame contains a 4-bit function code which indicates the command and the size of the expected slave frame, and a 12-bit field for address or parameters.

All devices on the bus decode the master frame. The addressed source device replies with its slave frame, which may be read by several other devices. Both the master frame and the slave frame are broadcasted.

Since data are identified by their source address only, the configuration tools subscribe each device (as a source or destination) to the periodic data it is interested in.

3 [part 1 / p 43]
4 [part 3 / p 43]

Frame format

The frame format is optimised for short and frequent data items:

- The master frame has a fixed size of 33 bits (9 + 16 + 8).
- Slave frames comprise a 9 bit frame delimiter, and a data portion which length depends on the function code in the master frame, followed by an 8-bit check sequence per 64 bit data. Longer slave frames use two or four 64+8-bit data-portions.

⁵⁾The expected length of the responding slave frame is:

Function-code	0	1	2	3	4
Data-length	16	32	64	128	256
Check sequences	1	1	1	2	4
Frame-length	33	49	81	153	297

Calculated throughput

⁶⁾For the repeaters a delay of 1.5 μ s is assumed. For the propagation delay 6.0 μ s/km are assumed for a loaded line. 4.0 μ s account for the decoding of the master frame and reply at the source. 1.6 μ s are assumed for the master's delay between the reception of a slave frame and sending a master frame.

The performance of the Multifunction Vehicle Bus is demonstrated by two extreme configurations:

- 30m bus with no repeater: 0.3 6 μ s propagation delay + 4 μ s source + 1.6 μ s master + frame duration.
- 2 km bus with 3 repeater links (with 2 regenerators each): 33 μ s propagation delay + 4 μ s source + 1. 6 μ s master + frame duration.

data	tele-gram	telegram dura-tion (without delays)	telegram rate [1/s]		net data rate [kb/s]	
bits	bits	[μ s]	30 m	2000 m	30 m	2000m
16	66	44.0	20'000	12'106	320	194
32	82	54.7	16'474	10'718	527	342
64	114	76.0	12'195	8'726	780	558
128	186	124.0	7'692	6'150	984	787
256	330	220.0	4'424	3'867	1'132	990

5 [part 3 / p 50]

6 [part 3 / p 48]

Ports on the MVB

A terminology issue

⁸⁾For the less initiated reader various terms meaning (nearly) the same might be confusing. By tradition the object of interest is named differently in the various levels of communication. According to the OSI model (see appendix), these are:

OSI level				Remarks
Process variable	User messages, system messages	Supervisory data	Application (level 7)	The application's view on the frame data
Dataset	Message transfer protocol		Transport (4)	Frame data (without check sums)
Periodic data	Message data	Supervisory data	Logical link control (2b)	
Ports (periodic access)	Queues (sporadic access)	Supervisory access	Medium access control (level 2a)	Frame data (without check sums) + control information
	Frames with logical address	Frames with physical address	Physical (1)	Data (function code, address, etc.) and check sums

Purpose of ports

⁹⁾ The interval between a master frame (received at the slave side) and the responding slave frame is less than 4 μ s. To allow a slave frame to respond within this time, the frame data is prepared for transmission beforehand. A port provides also a means to control access, data freshness and interrupts. Each device owns a certain number of ports, which are configured as source ports or as sink ports.

A port ensures that the data is read consistently when it is accessed simultaneously by the bus and by the application processor on the device. This requires at least two memory pages, which are used alternatively: while one page is updated by a data-producer (that is, the application for source-ports, the bus access processor for sink-ports), the other can be accessed.

Two kinds of ports exist: physical and logical ports.

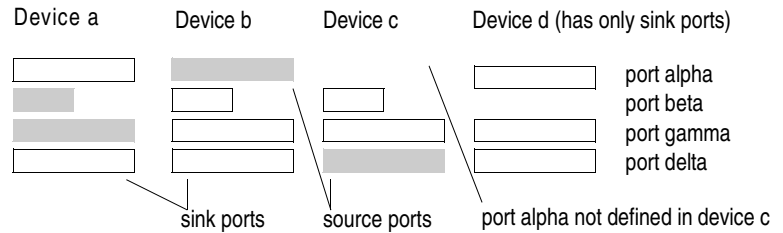
Logical port

Logical ports provide the basic communication for process data. They can be seen as windows to the data on the bus. At least two devices on the bus subscribe to a particular port on the bus. Only

8 [part 1 / p 30]

9 [part 1 / p 46 and 78]

one device may define a particular port as source. On all other devices this port is a sink (consumer of data).



The size of logical ports may be 16, 32, 64, 128 or 256 bits – the same as the data-lengths of slave frames.

A logical port is selected with function code 0..4 and the 12-bit logical address in the master frame (codes 5..7 are reserved). Each device implements a number of logical ports, typically 256. At configuration time the logical ports are configured either as source or as sink.

Physical port

A physical port is selected with function code 8..15 and the 12-bit device address in the master frame. Each device implements 8 physical ports which are used for supervisory data (16 bits) and message data (256 bits). Function codes 10 (supervisory data) and 11 (message data) are reserved.

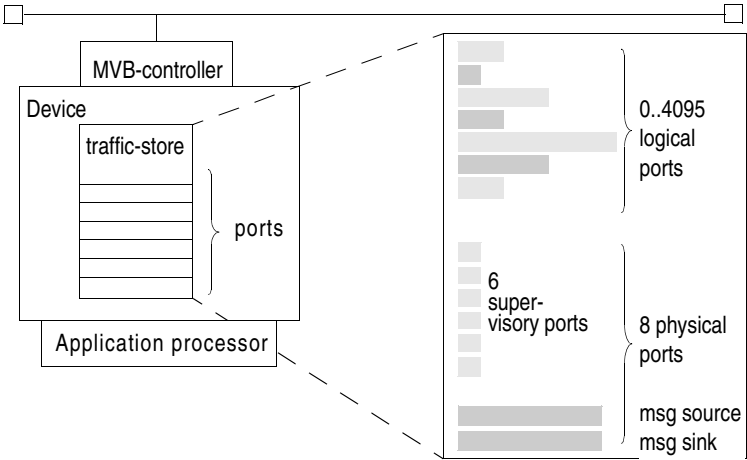
- | | |
|--------|--|
| 8 | Mastership transfer: addressed slave accepts/rejects. |
| 9 | Start event poll (12-bit field contains parameters, not an address): all bidding slave devices answer. |
| 12 | <p>Message poll: addressed slave answers with 256 bit data. All other devices will inspect the slave frame and accept it if their own address appears in the header of the slave frame.</p> <p>The contents of the message data port may not be over-written by the bus controller on the device. A frame is ignored if the previous contents of the message port has not been fetched by the application.</p> |
| 13, 14 | Group event poll and individual event poll: a group of slaves or a single slave answers. |
| 15 | Device status poll: Slave device answers with status. |

With the application Virtual Terminal the user addresses logical ports. Physical ports are addressed only by special functions, such as "Read Diagnostic".

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Ports in a device

The following diagram shows the ports in a device. They are located in the traffic store, a shared memory accessed both by the application processor and by the bus controller.



A port index table (PIT) maps the logical port addresses to the device specific layout of the traffic store. This allows to map a large address space to a small traffic store.

In addition to the logical and physical ports, the message queues may also be located in the traffic store. These are attached to the physical message port.

Freshness supervision

¹⁰⁾When a device fails, it ceases to produce new values. The sinks operate with the former value which becomes obsolete. This situation is uncovered by freshness supervision.

When the application reads a variable, the port indicates how long ago it has been updated. The application decides on the action to take.

Medium access

¹¹⁾The Multifunction Vehicle Bus is administered by a dedicated device, the bus administrator. The administrator is the only bus master, that is, the only device entitled to send master frames. All other devices are slaves, they may not transmit spontaneously. In addition to the master function the administrator provides services like clock, configuration, supervision or directory.

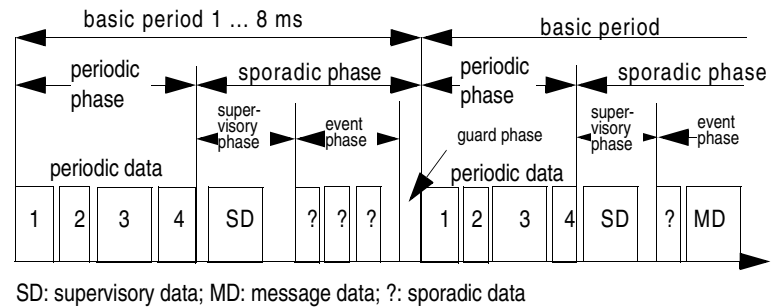
The master may be placed at any location in the bus. To increase availability, several bus administrators may exist. They transmit mastership by a token. At a given time, only one administrator is in charge of the bus.

Periodic and sporadic transmission

The bus master periodically polls the ports according to a predefined sequence (see *Periodic scan list* on page 16).

Messages are less time critical and thus transmitted sporadically, that is, on demand.

¹²⁾To serve both purposes the bus time is divided into a periodic phase and a sporadic phase:



Periodic phase

The periodic phase occupies a fixed portion of the traffic on the bus. During that time, the bus master polls the devices in sequence. A polled device answers with a frame containing its periodic data. This answer is broadcasted to all devices. Each device picks up from the frame the data items it is interested in. The format of this frame is fixed at initialisation time.

Periodic data do not need to be acknowledged by the destination devices, since these data is transmitted periodically. This traffic is also called source-addressed broadcast.

The periodic phase is reserved for transmission of process data.

11 [part 1 / p 48]
12 [part 3 v1.8 / p 69]

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Sporadic phase	Between the periodic phases the sporadic phases allow the devices to transmit data if they have some. This traffic is also called event-driven or on-demand traffic. When the bus master calls the devices for sporadic data, only devices with data to transmit respond with a filled frame. Only message data is transmitted in the sporadic phase. It can be broken down further into the supervisory phase, the event phase and the guard phase.
Supervisory phase	The supervisory phase is reserved for scanning the status of the devices and for transfer of mastership.
Event phase	Events are any state change which cause a device to request transmission. Therefore, the traffic must be acknowledged to ensure that no state change gets lost.
Guard phase	The guard phase acts as a buffer after the sporadic transmissions (events and supervisory data) to allow an exact start of the next periodic phase. The master does not send a master frame if the transmission of the longest expected response would take longer than the time remaining until the next basic period is scheduled.

Bus mastering

The bus master *polls* the devices on the bus according to the entries in the *master frame table*. This table is also called *periodic scan list*, because it is worked off periodically. Master frame tables are part of a bus administrator list.

The master issues a master frame at least every 1.3 ms to allow each device to monitor the line.

Basic cycle time

To enable the bus both for 50Hz and 60Hz systems, the MVB Virtual Terminal and PCNode provides two settings of the basic cycle time: 1 ms or 0.833 ms. 1 ms is the value according to MVB standard and is used for any time-values in this chapter.

Basic period

The bus master divides its turn (see later) into fixed periods, called basic periods. The basic period T_{bp} may take a value of 1, 2, 4 or 8 ms (multiples of the basic cycle time). It is generally divided into the 4 phases:

- periodic phase
- supervisory phase
- event phase
- guard phase

Individual poll period

Each periodic data source is polled with its individual period T_{ip} . This period is defined by the application. The individual poll period is equal to the basic period multiplied by a power of 2, but can not exceed 1024 ms. Ports with the same individual poll period belong to the same cycle¹³).

Cycle

A cycle groups periodic data with the same individual poll period. It gets the same name as its period in multiple of the basic period. For example, *cycle_4* is polled every 8 ms when the basic period is 2 ms.

A cycle may be split over several basic periods into sub-cycles. Cycles are named according to the number of sub-cycles. Hence *cycle_4* has 4 sub-cycles. The sub-cycles are indexed, starting with 0. For example, the second sub-cycle of *cycle_4* has name 4.1.

Macro period

The longest individual period in the periodic scan list is called the macroperiod. It can not exceed 1024 ms.

Macro cycle

A macrocycle is the number of cycles which fills a macro period. Therefore it lasts $1024/T_{bp}$ basic periods. T_{bp} is the length of the

¹³ Definition from [part 3 v 1.8 / p 72]; in [part 1 / p 48] this is called group.

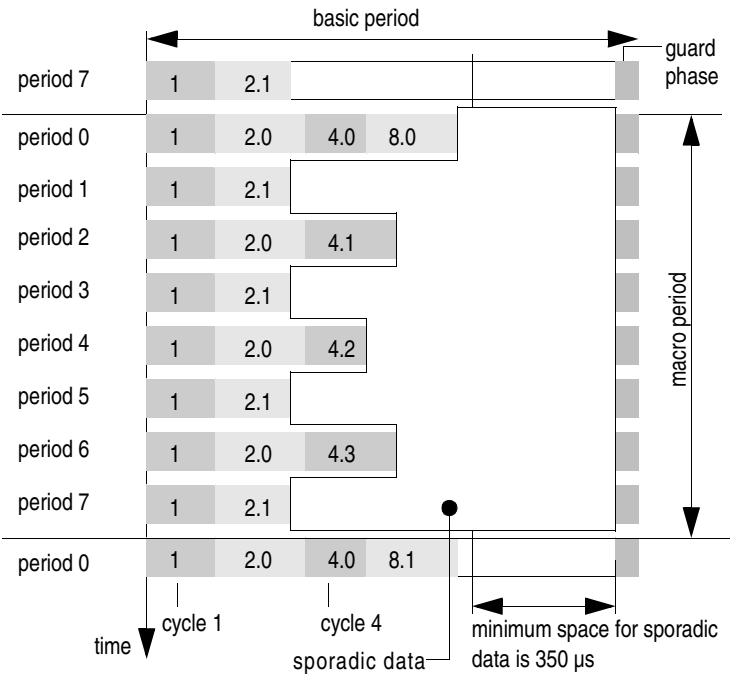
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basic period in ms. For example, for $T_{bp} = 4\text{ ms}$ a macro cycle lasts 256 basic periods.

Turn
Medium access is controlled by a central bus master, which remains in control of the bus for the duration of a turn.
A turn is defined by the number of macrocycles after which bus mastership must be transferred. A supervisory command may terminate a turn after each macrocycle. If no other bus administrator is available, the master remains the same.
The beginning of a turn may be used to synchronise devices.

Transfer of mastership
The sporadic phase in the last period of a turn is used for transfer of bus mastership.

Periodic scan list
The periodic scan list defines the addresses sent out periodically and the time left for the sporadic phase in each basic period of a turn.
The following figure shows the construction of a macrocycle consisting of 8 cycles as an example. Real macrocycles have 128 to 1024 cycles.¹⁴⁾



14 numbering of cycles and sub-cycles is inconsistent between [part 1 / p 49] and [part 3 v 1.8 / p 73]. If [par 3 v 1.8 / p 72] is correct, then there must (in my humble opinion) exist 4.2 and 4.3.

Process data

Traffic store

¹⁵⁾The ports of a device are located in the traffic store, a shared memory which can be accessed both by the application (in particular, the Virtual Terminal application of the MVB Virtual Terminal and PCNode) and by a bus controller. In very simple devices (field device) the ports are reduced to a register, there is no traffic store.

¹⁶⁾Devices may have up to 16 traffic stores. These need not have the same structure, although this is advantageous. This permits application programs residing in the same device to communicate in the same way as if there were located in different devices. The devices handled by Virtual Terminal may have only one traffic store.

Dataset

¹⁷⁾The contents of a port is a dataset. A dataset is just a stream of bits, which need interpretation according to the requirements of the application. All or none data of a dataset are transmitted. A dataset is static: its format and size may not be modified on-line.

The different process data of a dataset may be consumed by different devices, but a dataset is produced by only one device. When polled by the bus master, the producer device broadcasts a frame containing the dataset. This frame is received by all devices subscribed to the port carrying that dataset. Each device picks from the dataset the process data it is interested in.

The format of datasets – its interpretation – is fixed by the hardware for field devices and configured by the ‘programming and test station’ for complex devices (for example, the MVB Virtual Terminal and PCNode). The configuration is loaded during initialisation time and does not change during operation.

Process variables

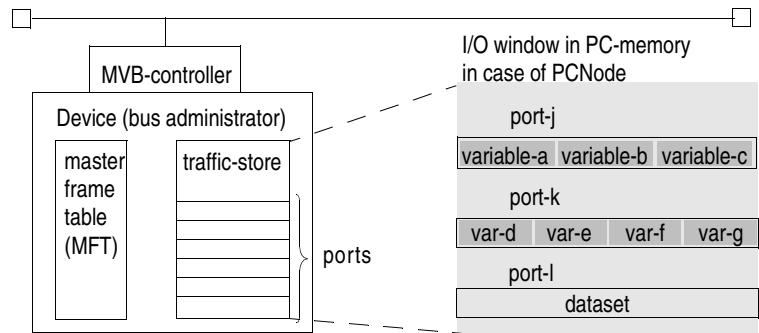
A dataset is interpreted by process variables. Within a dataset, a variable is identified by its bit offset. A dataset is just a stream of bits, whereas process variables have a type, for example integer, or

15 [part 1 / p 80]

16 [part 1 / p 88]

17 [part 1 / p 82]

character. A dataset can be interpreted by more than one process variables, but their definitions must not overlap.



The dataset of a sink port need not be defined completely by process variables. The application on that device might ‘pick up’ only the desired information.

Depending on the computer architecture used for the device (for example, the PCNode of the MVB Virtual Terminal and PCNode) bit-oriented variables and word-oriented variables can not be mixed in one dataset. Otherwise the MVB controller can not resolve the difference in bit-ordering of little-endian systems (Intel, DEC) and big-endian systems (Motorola, IBM mainframe).

Variable types

¹⁸⁾While the lower layers of the transport protocol ensure that a bit stream can be sent from one device to another, the presentation layer ensures that the communicating parties agree on the meaning of that bit-stream. When a device produces a variable, the consumers must know the type to interpret the value. However, the specific meaning of a variable is only known when considered in context.

Example: a transducer manufacturer specifies that its wheel speed is transmitted using the FRACTIONAL200% format. With this indication, the destinations of the variables know that the speed is, for instance, 87.5% of nominal speed. The value of the nominal speed is an additional convention which is not part of the presentation layer, but the application must know about it.

Basic and constructed types

In all modern computer languages, a variable is defined before a program uses it. This declaration defines the type of the variable. There are basic (built into the language) types like “integer”, “float” or “character”. Structured types are assemblies of basic types and may be structured as “array”, “record”, “set” or “file” of basic types. The storage in memory of a variable of a specific

18 [part 1 / p 90] and [part 2 / p 143]